COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	2026-08-29	**DATE SUBMITTED:**	2026-08-30
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ACTIVITY NO. 5.1 **Implement Ansible roles in playbooks**

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

Issue the command git pull

What happened?

```
localserver@workstation:~/CPE232_Duenas$ git pull
Already up to date.
```

- On the last activity i already did the git pull origin main

Edit install_apache.yaml

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
    when: ansible_ distribution == "Ubuntu"
```

ansible-playbook –ask-become-pass install_apache.yaml

```
PLAY RECAP *********************************************************************************
centos1                    : ok=1    changed=0    unreachable=0    failed=0    skipped=3    resc
ued=0    ignored=0
ubuntu1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=0    resc
ued=0    ignored=0
ubuntu2                    : ok=4    changed=0    unreachable=0    failed=0    skipped=0    resc
ued=0    ignored=0

localserver@workstation:~/CPE232_Duenas/ansible-lab$
```

Edit install_apache.yaml

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"
```

```
  - name: update repository index
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install apache2 package
    dnf:
      name: httpd
      state: latest
    when: ansible_distribution == "CentOS"

  - name: add PHP support for apache
    dnf:
      name: php
      state: latest
    when: ansible_distribution == "CentOS"
```

ansible-playbook –ask-become-pass install_apache.yaml

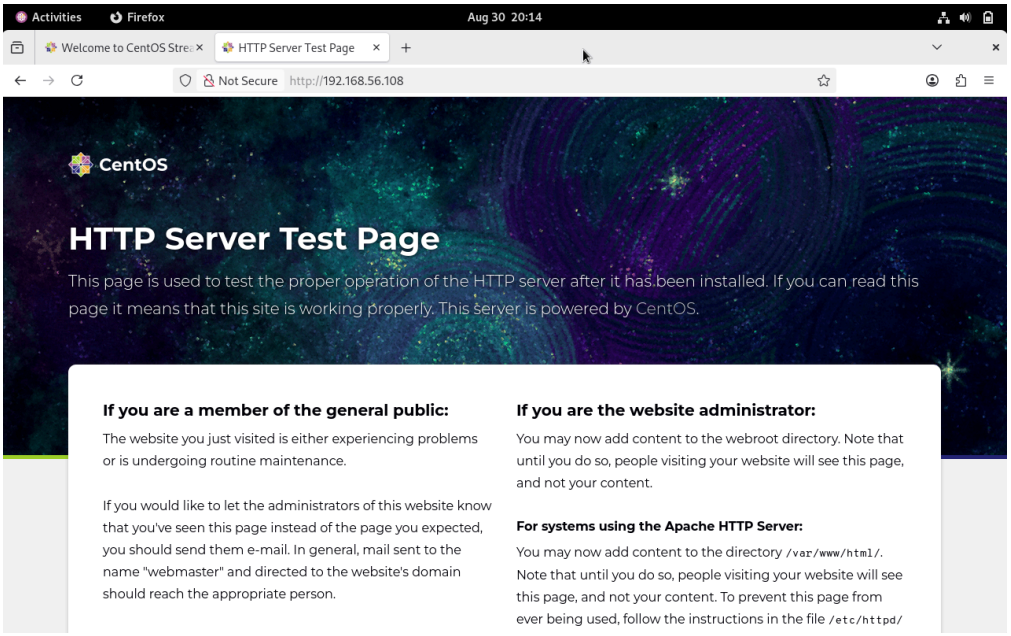
```
PLAY RECAP *********************************************************************************
centos1                    : ok=4    changed=2    unreachable=0    failed=0    skipped=3    resc
ued=0    ignored=0
ubuntu1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=3    resc
ued=0    ignored=0
ubuntu2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=3    resc
ued=0    ignored=0

localserver@workstation:~/CPE232_Duenas/ansible-lab$
```

- What happened was the playbook executed successfully across all target hosts using apt for Ubuntu and dnf for CentOS. CentOS recorded two changed tasks, while both Ubuntu hosts skipped the CentOS-specific tasks.

Activating apache http server on CentOS

```
[centos1@linux ~]$ systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: d>
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: inactive (dead)
       Docs: man:httpd.service(8)
lines 1-6/6 (END)
[1]+  Stopped                 systemctl status httpd
[centos1@linux ~]$ sudo systemctl start httpd
[sudo] password for centos1:
[centos1@linux ~]$ sudo firewall-cmd --add-port=80/tcp
success
[centos1@linux ~]$
```



- I successfully made apache httpd server work on CentOS

Task 2: Refactoring playbook

Edit install_apache.yaml

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index Ubuntu
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 and php packages for Ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"
```

```
- name: update repository index for CentOS
  dnf:
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: install apache and php packages on CentOS
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"
```

ansible-playbook –ask-become-pass install_apache.yaml

```
PLAY RECAP *********************************************************************************
centos1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    resc
ued=0    ignored=0
ubuntu1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    resc
ued=0    ignored=0
ubuntu2                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    resc
ued=0    ignored=0

localserver@workstation:~/CPE232_Duenas/ansible-lab$
```

- What happened was the playbook ran cleanly across all target hosts with zero failures or changes recorded. Non-matching tasks were skipped appropriately based on the specified distribution conditions.

Edit install_apache.yaml

```
---
- hosts: all
  become: true
  tasks:

#  - name: update repository index Ubuntu
#    apt:
#      update_cache: yes
#    when: ansible_distribution == "Ubuntu"

  - name: install apache2 and php packages for Ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
    when: ansible_distribution == "Ubuntu"
```

```
  - name: install apache and php packages on CentOS
    dnf:
      name:
        - httpd
        - php
      state: latest
      update_cache: yes
    when: ansible_distribution == "CentOS"
```

ansible-playbook –ask-become-pass install_apache.yaml

```
PLAY RECAP *********************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    resc
ued=0    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    resc
ued=0    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    resc
ued=0    ignored=0
```

- What happened was the play tasks executed successfully with lower overall task counts per host. The skipped task counts dropped from two down to one because separate repository index tasks were removed.

Edit install_apache.yaml

```
---
- hosts: all
  become: true
  tasks:

#  - name: update repository index Ubuntu
#    apt:
#      update_cache: yes
#    when: ansible_distribution == "Ubuntu"

  - name: install apache and php # packages for Ubuntu
    apt:
      name:
        - " {{ apache_package }} "
        - " {{ php_package }} "
      state: latest
      update_cache: yes
#    when: ansible_distribution == "Ubuntu"
```

ansible-playbook –ask-become-pass install_apache.yaml

```
PLAY RECAP *********************************************************************************
centos1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    resc
ued=0    ignored=0
ubuntu1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    resc
ued=0    ignored=0
ubuntu2                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    resc
ued=0    ignored=0
```

- What happened was the playbook execution failed on all target hosts. This occurred because the required package variables were not defined yet, and the apt module was hardcoded for all systems including CentOS.

Edit install_apache.yaml

```
all:
  children:

    ubuntu:
      vars:
        ansible_user: gab
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.113
          ansible_user: ubuntu1
          ansible_ssh_private_key_file: ~/.ssh/id_rsa
          ansible_password: "123"
          os_type: Ubuntu

          # update here
          apache_package: apache2
          php_package: libapache2-mod-php

        ubuntu2:
          ansible_host: 192.168.56.114
          ansible_user: ubuntu2
          ansible_ssh_private_key_file: ~/.ssh/id_rsa
          ansible_password: "123"
          os_type: Ubuntu

          # update here
          apache_package: apache2
          php_package: libapache2-mod-php
```

```
    centos:
      vars:
        ansible_user: ranz
      hosts:
        centos1:
          ansible_host: 192.168.56.108
          ansible_user: centos1
          ansible_ssh_private_key_file: ~/.ssh/id_rsa
          ansible_password: "123"
          os_type: Centos

          # update here
          apache_package: httpd
          php_package: php

    linuxservers:
      children:
        ubuntu:
        centos:
```

Edit install_apache.yaml

```
---
- hosts: all
  become: true
  tasks:

#  - name: update repository index Ubuntu
#    apt:
#      update_cache: yes
#    when: ansible_distribution == "Ubuntu"

  - name: install apache and php # packages for Ubuntu
    package:
      name:
        - " {{ apache_package }} "
        - " {{ php_package }} "
      state: latest
      update_cache: yes
#    when: ansible_distribution == "Ubuntu"
```

ansible-playbook –ask-become-pass install_apache.yaml

```
PLAY RECAP *********************************************************************************************
centos1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    resc
ued=0    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    resc
ued=0    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    resc
ued=0    ignored=0
```

- What happened was the playbook completed successfully on all hosts without any errors. The generic package module correctly applied the appropriate package manager and variables defined in the inventory for both OS types.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a playbook that could do the previous tasks in Red Hat OS.

```
[centos1@linux CentOS-ansible-lab]$ ls
ansible.cfg  install_apache_centos.yaml  inventory.yaml
```

a. Creating an inventory.yaml

```
centos:
  vars:
    ansible_user: ranz
  hosts:
    centos1:
      ansible_host: 192.168.56.108
      ansible_user: centos1
      ansible_ssh_private_key_file: ~/.ssh/id_rsa
      ansible_password: "123"
      os_type: Centos

      # update here
      apache_package: httpd
      php_package: php

linuxservers:
  children:
    #       ubuntu:
    centos:
```

b. Creating an ansible.cfg

```
[defaults]
inventory = inventory.yaml
```

c. Creating an install_apache_centos.yaml

```yaml
---
- name: Configure Apache and PHP on CentOS
  hosts: centos
  become: true
  tasks:

    - name: Install Apache and PHP packages
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes

    - name: Start and Enable apache service
      service:
        name: httpd
        state: started
        enabled: yes

    - name: Allow firewalld
      firewalld:
        service: http
        permanent: yes
        state: enabled
        immediate: yes
```

d. Result

```
[centos1@linux CentOS-ansible-lab]$ ansible-playbook --ask-become-pass install_a
pache_centos.yaml
BECOME password:
[WARNING]: Collection ansible.posix does not support Ansible version 2.14.18

PLAY [Configure Apache and PHP on CentOS] ***************************************

TASK [Gathering Facts] **********************************************************
ok: [centos1]

TASK [Install Apache and PHP packages] ******************************************
ok: [centos1]

TASK [Start and Enable apache service] ******************************************
changed: [centos1]

TASK [Allow firewalld] **********************************************************
changed: [centos1]

PLAY RECAP **********************************************************************
centos1                    : ok=4    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

[centos1@linux CentOS-ansible-lab]$
```

- What happened was the playbook successfully installed Apache and PHP using the dnf module, enabled and started the httpd service, and configured firewalld to allow HTTP traffic. The execution completed with two changed tasks and no errors on the CentOS host.

Reflections:

Answer the following:

1. Why do you think refactoring of playbook codes is important?
   - Refactoring reduces duplicate code and makes playbooks easier to maintain, read, and execute efficiently. It allows playbooks to run faster by making tasks into fewer, cleaner instructions.

2. When do we use the “when” command in playbook?
   - The when command is used to run specific tasks only if defined conditional statements are met. For example, it allows a task to execute on certain operating systems while automatically skipping others.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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